In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE(Main)-2017

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – IV
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	A
2.	B	C	C
3.	B	B	C
4.	A	A	A
5.	B	A	A
6.	A	A	D
7.	B	B	B
8.	B	B	C
9.	C	B	B
10.	D	D	B
11.	C	B	B
12.	A	A	B
13.	C	A	A
14.	A	B	B
15.	D	B	A
16.	B	C	D
17.	D	B	A
18.	B	B	B
19.	D	A	A
20.	C	B	A
21.	D	D	A
22.	A	C	A
23.	D	B	C
24.	C	A	B
25.	C	B	A
26.	B	C	A
27.	D	D	A
28.	C	D	C
29.	C	A	D
30.	A	D	D

Physics

PART – I

SECTION – A

1. **C**

The x-coordinate is obtained by integrating the expression for v_x, and the x-component of the acceleration is obtained by differentiating v_x. Thus,

$$\int dx = \int vx\ dt \qquad \int_0^x dx = \int_0^x (50-16t)dt \qquad x = 50t - 8t^2\ m$$

$$a_x = \frac{d}{dt}(50-16t) \qquad a_x = -16\,m/s^2$$

The y-components of velocity and acceleration are

$$\left[v_y - \bar{y}\right] \qquad v_y = \frac{d}{dt}(100-4t^2) \qquad v_y = -8t\ m/s$$

$$\left[a^y - v_y\right] \qquad a_y = \frac{d}{dt}(-8t) \qquad a_y = -8m/s^2$$

We now calculate corresponding values of x and y for various values of t and plot x against y to obtain the path as shown.

When $y = 0,\ 0 = 100 - 4t^2$, so $t = 5\,s$. For this value of the time, we have

$$v_x = 50 - 16(5) = -30\ m/s$$

$$v_y = -8(5) = -40\ m/s$$

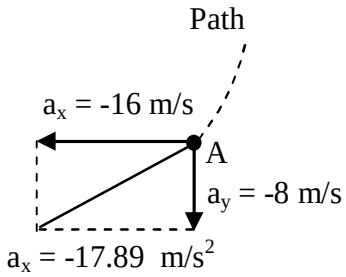
$$v = \sqrt{(-30)^2 + (40)^2} = 50 m/s$$

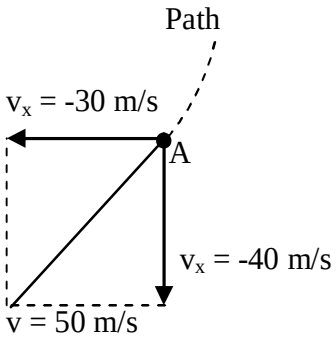
$$a = \sqrt{(-16)^2 + (-8)^2} = 17.89\ m/s^2$$

The velocity and acceleration components and their resultants are shown on the separate diagram for point A, where y = 0. Thus, for this condition we may write

$$v = -30\hat{i} - 40\hat{j}\ m/s$$

$$a = -16\hat{i} - 8\hat{j}\ m/s^2$$





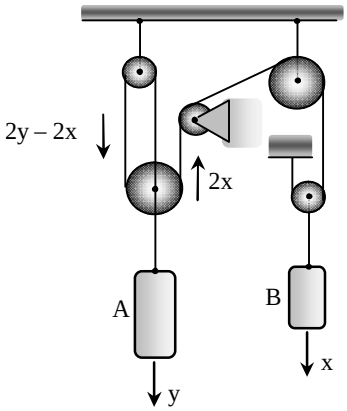
2. **B**

$$v_x = (v_0 \cos\theta)e^{-kt}$$

$$v_y = \left(v_0 \sin\theta + \frac{g}{k}\right)e^{-kt} - \frac{g}{k}$$

$$v_x \to 0,\ \ v_y \to -\frac{g}{k}$$

3. **B**



From figure,

$2x - 2y = y$

$x = \frac{3y}{2}$

$v_x = \frac{3}{2}v_y$

$a_x = \frac{3}{2}a_y$

$v_B = 1.2\ m/s$ up

$a_B = 3\ m/s^2$ down

4. **A**

We express the mean kinetic energy of a helium molecule in terms of the temperature: $\bar{E}_k = (3/2)kT_0$. For monatomic gases like helium, this will be the total kinetic energy of molecules:

$$\bar{E} = \frac{3}{2} \times 1.38 \times 10^{-23}\ J/K \times 273K \simeq 5.65 \times 10^{-21} J.$$

Since $\bar{E} = m_0 \bar{v}_{rms}^2/2$, we have $\bar{v}_{rms} = \sqrt{2\bar{E}/m_0}$, where m_0 is the mass of a helium molecule, which can be expressed as the ratio of the molar mass and the Avogadro constant: $m_0 = M/N_A$.

Finally, we obtain

$$\bar{v}_{rms} = \sqrt{\frac{2\bar{E}N_A}{M}},$$

$$\bar{v}_{rms} = \sqrt{\frac{2 \times 5.65 \times 10^{-21} J \times 6.02 \times 10^{23} mol^{-1}}{4 \times 10^{-3} kg/mol}} \simeq 1300\ m/s$$

5. **B**

Voltage across load resistances $V_2 = V_1 - 2iR$

$$= 150 - 2 \times 1.7 \times 10^{-3} \times \frac{650 \times 10^3 \times 1.2 \times 10^{-7}}{5 \times 10^{-6}}$$

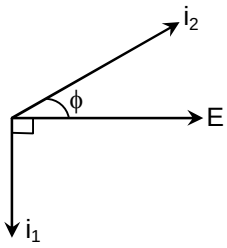
V_2 = 97 volt

6. **A**

$$X_L = \omega L = 40\pi \times \frac{1}{40\pi} = 1\ \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{40\pi}\ \Omega$$

$R = 1\ \Omega$



$\tan\phi = \frac{X_L}{R} = 1$

$\phi = 45^o$

$X_C = \sqrt{2}\ \Omega$

$X_L = 1\,\Omega,\ R = 1\,\Omega$

$z = 1\,\Omega$

7. **B**

We use ε to denote the angle formed by the direction of incident rays with the vertical wall of the vessel. It can be seen from the figure that the angle of incidence ε and the altitude of the Sun φ are related: $\varepsilon + \varphi = \pi/2$, $\varepsilon = 90^o - 60^o = 30^o$.

The rays propagate in the empty vessel along a straight line (along AM), and the length *l* of the umbra can be calculated from the right-angled triangle AMD:

$l_1 = AD = H \tan\ \varepsilon,\ \tan 30^o = 0.577,$

$l_1 = 25\ \text{cm} \times 0.577 \simeq 14.4\ \text{cm}.$

If the vessel contains water, he rays will be refracted at the air-water interface since a light ray bends away from the rarer medium into a denser medium. In this case, the refracted ray is closer to the vertical, and the angle of refraction ε' is smaller than the angle of incidence ε. The length of the umbra in the vessel filled with water is

$l_2 = BD = BC + CD.$ But $CD = ON.$

In the right-angled triangle OMN, we have ON = (H –h) $\times$ tan ε, ON = 5 cm $\times$ 0.577 $\simeq$ 2.9 cm. From the triangle BOC, we find BC = h tan ε'. In order to determine the angle of refraction, we use the second law of refraction: sin ε/sin ε' = n, sin ε' = sin ε/n, sin ε' = 0.5/1.33 = 0.3759, so tan ε' = 0.404. Consequently, BC = 20 cm $\times$ 0.404 = 8.1 cm.

The length of the umbra in the vessel with water is

$l_2 = BD = 8.1\,\text{cm} + 2.9\ \text{cm} = 11\text{cm}.$

When the vessel is filled with water, the length of the umbra decreases by Δl:

$\Delta l = 14.4\ \text{cm} - 11\,\text{cm} = 3.4\ \text{cm}.$

8. **B**

In order answer the question, we must know the path difference for the rays. The optical path difference is equal to their geometrical difference (since the rays propagate in the same medium, viz. air):

$\Delta = S_2D = S_2D - S_1O,\ S_1O = l.$

From the triangle S_1, OS_2, we determine $S_2O : S_2O = \sqrt{l^2 + d^2} = l\sqrt{1 + (d/l)^2}$. since d/l is much smaller than l, we can use approximation $\left(\sqrt{1 \pm a^2} = 1 \pm \frac{1}{2}a^2\right)$, whence $S_2 = O = l\left[1 + \frac{1}{2}\left(\frac{d}{l}\right)^2\right]$.

Then

$\Delta = l\left[1 + \frac{1}{2}\frac{d^2}{l^2} - 1\right] = \frac{d^2}{2l},$

$$\Delta = \frac{\left(2.2\times10^{-3}\text{m}\right)^2}{2\times2.2\text{ m}} = 1.1\times10^{-6}\text{m}.$$

There will be constructive interference at O if the path difference contains integral number of waves, i.e. k =1, 2, 3, …….:

$$k = \frac{\Delta}{\lambda} = \frac{1.1\times10^{-5}\text{m}}{5.5\times10^{-7}\text{m}} = 2$$

Answer. Constructive interference (bright fringe) occurs at O.

9. **C**

$$\omega^2 = \omega_0^2 - \left(\frac{b}{2m}\right)^2$$

$$= \frac{k}{m} - \left(\frac{b}{2m}\right)^2$$

$$= \frac{100}{1} - \frac{100}{4}$$

$$\omega = 5\sqrt{3} = 8.5\ \text{s}^{-1}$$

10. **D**

The total time interval is the sum of the two time intervals.
In each wire

$$\Delta t = \frac{L}{v} = L\sqrt{\frac{\mu}{T}}$$

Let A represent the cross-sectional area of one wire. The mass of one wire can be written both as $m = \rho V = \rho AL$ and also as $m = \mu L$.

Then we have $\mu = \rho A = \frac{\pi\rho d^2}{4}$.

Thus, $\Delta t = L\left(\frac{\pi\rho d^2}{4T}\right)^{1/2}$

For copper,

$$\Delta t = \left(20.0\text{m}\right)\left[\frac{\left(\pi\right)\left(8920\text{ kg/m}^3\right)^2}{4T}\right]^{1/2} = 0.137\text{ s}$$

For steel,

$$\Delta t = \left(30.0\text{ m}\right)\left[\frac{\left(\pi\right)\left(7860\text{ kg/m}^3\right)\left(1.00\times10^{-3}\right)^2}{\left(4\right)\left(150\text{ N}\right)}\right]^{1/2} = 0.192\text{ s}$$

The total time interval is 0.137+0.192 = 0.329 s

11. **C**

According to Equation 38.1, dark bands (minima) occur where

$$\sin\theta = m\frac{\lambda}{a}$$

For the first minimum, m = 1, and the distance from the center of the central maximum is

$$y_1 = L\tan\theta \approx L\sin\theta = L\left(\frac{\lambda}{a}\right)$$

Thus, the needed distance to the screen is

$$L = y_1\left(\frac{a}{\lambda}\right) = \left(0.85\times10^{-3}\right)\left(\frac{0.75\times10^{-3}\,m}{587.5\times10^{-9}\,m}\right) = 1.1\,m$$

12. **A**

The allowed energy levels of the hydrogen atom are given by

$$E_n = -\frac{13.6\,eV}{n^2} \text{ where } n = 1, 2, 3.....$$

A transition in which a lower state n_i absorbs a photon of energy ΔE results in a higher state n_f, and energy is conserved:

$$E_1 + \Delta E = E_f$$

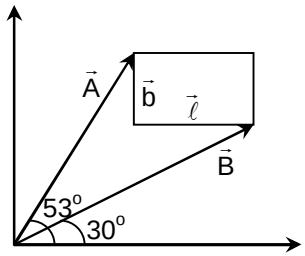
Or

$$\Delta E = E_f - E_i = -\frac{13.6\,eV}{n_1^2} = -\left(\frac{13.6\,eV}{n_1^2}\right) = 13.6\ eV\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

For the transition $n_i = 2$ to $n_f = 5$,

$$\Delta E = 13.6\,eV\left(\frac{1}{2^2} - \frac{1}{5^2}\right) = 2.86\ eV$$

13. **C**



$$\vec{A} + \vec{b} + \vec{\ell} = \vec{B}$$

$$10\left(\cos53\hat{i} + \sin53\hat{j}\right) + \left(-b\hat{j}\right) + \ell\hat{i} = 12\left(\cos30\hat{i} + \sin30\hat{j}\right)$$

$$6\hat{i} + 8\hat{j} + \ell\hat{i} - b\hat{j} = 6\sqrt{3}\hat{i} + 6\hat{j}$$

$$\ell + 6 = 6\sqrt{3}$$

$$8 - b = 6$$

$$b = 2\ m$$

$$\ell = 6\left(\sqrt{3} - 1\right) = 4.2\ m$$

14. **A**

The density of lead is $1.13\times10^4\,kg/m^3$, so we should expect our calculated value to be close to this value. The density of water is $1.00\times10^3\,kg/m^3$, so we see that lead is about 11 times denser than water, which agrees with our experience that lead sinks.

Density is defined as $\rho = m/V$. We must convert to SI units in the calculation.

$$\rho = \left(\frac{23.94\,g}{2.10\,cm^3}\right)\left(\frac{1kg}{1000\ g}\right)\left(\frac{100\ cm}{1m}\right)^3$$

$$= \left(\frac{23.94\,g}{2.10\ cm^3}\right)\left(\frac{1\,kg}{1000\,g}\right)\left(\frac{1000\ 000\,cm^3}{1\,m^3}\right)$$

$$= 1.14\times10^4\ kg/m^3$$

15. **D**
Conceptual.

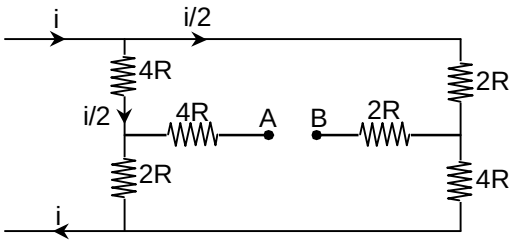
16. **B**
Conceptual.

17. **D**
Apply $v = u + at$.

18. **B**
Conceptual.

19. **D**
Number of photon $= \dfrac{20}{hc/\lambda}$

20. **C**



$V_A + 4R.\dfrac{i}{2} - 2R.\dfrac{i}{2} = V_B$

$V_A - V_B = iR$

21. **D**
In head on elastic collision between identical bodies velocities get interchanged.

22. **A**
Slope of graph $= \dfrac{1}{R}$

23. **D**
Ignore air resistance, the gpe reduces to 80% so 20% of the energy is lost.

24. **C**
$\dfrac{1}{2}mv^2 = \dfrac{1}{2} \times 5 \times 10^{-26} \times (500)^2 = 39$

25. **C**
2 minutes in 24 hours is 1/720 of a full rotation of the earth. So the angle subtended by the moon at the eyes is about ½ ^{0}. The distance earth – moon is a fixed value so the angle subtended by the earth at the moon is greater by 6400/1700. Hence 0.5 × 6400/1700 = 1.9^0.

26. **B**

Volume of the water of the film is fixed, so if the bubble has three times the radius, it has nine times the surface are, and as the film is very thin, this makes it thinner by a factor of nine in order to keep the volume of the film the same. This does expect that the thickness of the film t << R, the radius of the bubble.

27. **D**

A ratio question. The shapes are the same, and the ratio of the linear dimensions is 690/6.4 = 108, so the ratio of volumes is $\approx 10^6$.

28. **C**

The liquid has to flow faster through the narrower section because if unit volume passes a one point in the pipe every second, that same volume must flow through any other point each second otherwise there would be an accumulation of an incompressible fluid in the pipe.

29. **C**

The power dissipates in a resistor is given by I^2R. If R_A is unit current through unit resistance, R_B has twice the resistance and half the current, so that I^2R will be half the value (1/2W). R_C takes 3/2 units of current and is twice unit resistance sod resistance so dissipates 9/2 w. powers add up and so the total of these is 6W. The cell would supply the appropriate4 emf to supply these currents, otherwise the initial assumption of the 1W that we are given in R_A would not be right.

30. **A**

The rock only has to be lifted up to the neutral point and it will fall the rest of the way to earth.

Chemistry

PART – II

SECTION – A

1. **A**

By statement given in (a)

Wt. of NaCl = a g

Wt. of MCl = b g

$\therefore$ a + b = 1

Also, Meq. Of NaCl + Meq. Of MCl = Meq. of $AgNO_3$

= Meq. of AgCl

$= \frac{2.567}{143.5}\times 1000$

$\therefore$ Meq. of NaCl + Meq. of MCl = 17.889

Now, as reported in statement(b), mixture is heated to 300^oC, Since, NaCl is stable, the possibility seems for the decomposition of MCl and thus, after heating only MCl form vapours which react with $AgNO_3$

$\therefore$ Meq. of MCl = meq. of $AgNO_3$ = Meq. of AgCl

$= \frac{1.341}{143.5}\times 1000$

Meq. of MCl = 9.345

$\therefore$ Meq. of NaCl = 17.889 – 9.345 = 8.544

$$\frac{w}{58.5}\times 1000 = 8.544$$

$\therefore$ $w_{NaCl} = 0.50g$

$\therefore$ Meq. of MCl in mixture = 9.345

$\therefore$ $\frac{w}{M/1}\times 1000 = 9.345$

$\therefore$ $\frac{1-0.50}{M}\times 1000 = 9.345$ $(\because a+b=1)$

$\therefore$ M = 53.50

MCl may be NH_4Cl.

2. **C**

Given, for N_2 : Initially, P_1 = 2atm.

w_1 = 14 kg (diffusion time 1 hr)

After diffusion, $P_2 = \frac{1}{2}\text{atm},\ w_2 = ?$

Since, at constant V and temperature, for a gas

$P \propto w$

$\therefore$ $\frac{P_1}{P_2} = \frac{w_1}{w_2}$ or $\frac{2}{1/2} = \frac{14}{w_2}$

$\therefore$ $w_2 = \frac{14}{4}\text{kg}N_2$

$\therefore$ Weight of N_2 diffused in 1 hr = 14 – (14/4) = 21/2) kg

Similarly for H_2 : initially P_1 = atm, w_1 = 1 kg.

(diffusion time t hr)

After diffusion, $P_2 = \frac{1}{2}\text{atm}, \ w_2 = ?$

Again, $\frac{P_1}{P_2} = \frac{w_1}{w_2}$ or $\frac{2}{1/2} = \frac{1}{w_2}$

$\therefore \quad w_2 = \frac{1}{4}\text{kg}$

$\therefore$ Weight of H_2 and H_2 diffused $= 1 - \left(\frac{1}{4}\right) = \frac{3}{4}\text{kg}$

For diffusion of N_2 and H_2: t_{H_2} = 1 hr = 60 minute

$$\frac{w_{H_2}}{t_{H_2}} \times \frac{t_{N_2}}{w_{N_2}} = \sqrt{\frac{M_{H_2}}{M_{N_2}}}$$

$$\frac{(3/4)}{(21/2)} \times \frac{60}{t} = \sqrt{\frac{2}{28}}$$

t = 16 minute

3. **B**

Let work function of A and B be W_A and W_B respectively the,

$4.25 = W_A + T_A$

$T_A = 4.25 - W_A$ ....(i)

$T_B = 4.70 - W_B$ ....(ii)

$T_B - T_A = 0.45 + W_A - W_B$

$\because \quad T_B - T_A = -1.5$

$\therefore \quad W_B - T_A = 1.95$ ...(iii)

Now, $\lambda = \frac{h}{mu} = \sqrt{\frac{h}{2k.m}}$ $\left(\because \text{K.E.} = \frac{1}{2}mu^2\right)$

$\lambda \propto \frac{1}{\sqrt{k}}$ (K is kinetic energy)

$\frac{\lambda_B}{\lambda_A} = \sqrt{\frac{K_A}{K_B}} = 2$

$\therefore \quad \frac{K_A}{K_B} = 4 = \frac{T_A}{T_B}$ $(\because \text{K.E.} = T)$

$\frac{T_A}{T_A - 1.5} = 4$

$\therefore \quad T_A = 2\text{eV}$

$T_B = 0.5\text{eV}$

$W_A = 2.25\text{eV}$

$W_B = 4.2\text{eV}$

4. **A**

${}^{m}_{z}A \xrightarrow{-a} {}^{m-4}_{z-2}B \xrightarrow{-\beta} {}^{m-4}_{z-1}C$

$(t_{1/2})_A = 10\,\text{month} \quad (t_{1/2})_B = 6\,\text{month}$

The radioactive equilibrium is attached then, at equilibrium the ratio of atoms of A and B left is

$$\frac{N_A}{N_B} = \frac{t_{1/2}A}{t_{1/2}B} = \frac{10}{6} = \mathbf{1.66}$$

If half-life of A = 6 month and B is 10 month, then since $t_{1/2}A < t_{1/2}B$ or $\lambda_A > \lambda_B$ and thus no equilibrium will be set.

5. **A**

(a) $\because$ Eq. of O_2 = Eq.of Ag

or $w_{O_2}/8 = 1$ ($\because$ 1 mole of Ag = 1eq. of Ag)

$\therefore$ $w_{O_2} = 8g$

Using PV =(w/m) RT for volume of O_2, at T = 298 K, P = 750/760 atm

$$\frac{750}{760} \times V_{O_2} = \frac{8}{32} \times 0.0821 \times 298$$

$\therefore$ $V_{O_2} = 6.20$ litre

(b) Eq. of O_2 = Eq. of Ag.

or $$\frac{w_{O_2}}{8} = \frac{8 \times 10^{22} \times 108}{6.023 \times 10^{23} \times 108} \qquad \left(\therefore w_{Ag} = \frac{8 \times 10^{22} \times 108}{6.023 \times 10^{23}}\right)$$

$\therefore$ $w_{O_2} = 1.063g$

Again using PV=(w/m) RT: $V_{O_2} = 0.823$ litre

6. **A**

Mole of Mg = $\frac{1}{24}$

These moles of Mg will be converted to Mg^+ and Mg^{2+}. Let a mole of Mg^+ are formed, then

$$a \times 740 + \left(\frac{1}{24} - a\right) \times 2190 = 50$$

$\therefore$ $$a = 0.02845$$

$\therefore$ $$\%of\ Mg^+ = \frac{0.02845}{1/24} \times 100 = 68.28$$

$$\%of\ Mg^+ = 31.72$$

7. **B**

The stability of carbonates increases with increasing electropositive character of metal.

8. **B**

For $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

Before association	1	0
After association	$1-\alpha$	$\alpha/2$

Where, α is degree of association.

$\therefore$ $$i = 1 - \alpha + \frac{\alpha}{2}$$

Given, w = 0.2 g, W=20 g, $\Delta T = 0.45$

$\because \quad \Delta T = \dfrac{1000 \times K_f \times w}{m \times W}\left(1-\alpha+\dfrac{\alpha}{2}\right)$

or $\quad 0.45 = \dfrac{1000 \times 5.12 \times 0.2}{20 \times 60}\left(1-\alpha+\dfrac{\alpha}{2}\right)$

or $\quad 1-\alpha+\dfrac{\alpha}{2} = 0.527$

or $\quad \alpha = 0.946 \quad or \quad 94.6\%$

9. **B**

$$\frac{T_f}{T_b} = \sqrt{\frac{E_{ab}}{E_{af}}} = \sqrt{\frac{420}{500}} = \sqrt{0.84}$$

10. **D**

($\because$ Moles $\propto$ volume at constant P and T)

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Initial moles	a	0
Moles at equilibrium	(a –*x*)	2*x*

Given original volume = (75/100) or existing volume at equilibrium

Or Initial moles = (75/100) $\times$ moles at equilibrium

$\therefore \quad a = \dfrac{75}{100}(a+x)$

or $\quad x = \dfrac{25}{75}a = 0.33\ a$

$\therefore$ degree of dissociation, $\alpha = \dfrac{x}{a}$

$$= \frac{0.33a}{a} = 0.33\ or\ 33\%$$

11. **B**

$$N_2O_4 \rightleftharpoons 2NO_2$$

Initial moles	1	0
Moles at equilibrium	1— α	2α

Where, α is degree of dissociation = 0.503

$\because \quad K_p = \dfrac{\left(n_{NO_2}\right)^2}{\left(n_{N_2O_4}\right)} \times \left(\dfrac{p}{\sum n}\right)^{\Delta n}$

$$= \frac{(2\alpha)^2}{(1-\alpha)} \times \left(\frac{p}{1+\alpha}\right)1$$

$$Kp = \frac{4\alpha^2 p}{\left(1-\alpha^2\right)} = \frac{4 \times (0.503)^2 \times 1}{\left[1-(0.503)^2\right]}$$

Again at pressure P, $N_2O_4 \rightleftharpoons 2NO_2$

Initial moles	1	0
Moles at equilibrium	1-x	2x

Given $\frac{N_2O_4}{NO_2} = \frac{1-x}{2x} = \frac{1}{8}$

$\therefore \quad x = \frac{8}{10} = 0.8$

Using again,

$$Kp = \frac{4x^2.P}{1-x^2}$$

or $\quad 1.355 = \frac{4\times(0.8)^2.P}{\left[1-(0.8)^2\right]}$

or $\quad P = 0.19\ atm$

12. **A**

$$N_2O_4 \; f \;\; 2NO_2$$

Pressure at equilibrium 0.28 1.1

$\therefore \quad K_p = \frac{\left(P'_{NO_2}\right)^2}{\left(P'_{N_2O_4}\right)} = \frac{(1.1)^2}{0.28} = 4.32\ atm$

If volume of container is doubled, i.e., pressure becomes half, the reaction will also proceed in the half, the reaction will also proceed in the direction where the reaction shows an increase in moles, i.e., decomposition of N_2O_4 is favoured;

$$N_2O_4 \hat{\ddagger} \hat{\dagger} \; 2NO_2$$

New pressure at equilibrium $\left[\frac{0.28}{2} - P\right] \quad \left[\frac{1.1}{2} + 2P\right]$

Where, N_2O_4 equivalent to pressure P is used up in doing so.

13. **A**

$$pOH = -\log K_b + \log\frac{[conjugate\ acid]}{[Base]}$$

$$5 = 4.7 + \log\frac{a}{b}$$

$$\frac{a}{b} = 2 \quad \therefore\ a = 2b \qquad .....(i)$$

Given, $\quad a + b = 0.6 \qquad .....(ii)$

By Eqs. (i) and (ii), 2b+ b = 0.6

$\therefore \quad 3b = 0.6$

or $\quad b = 0.2\ mole = 0.2\times 17 = 3.4g\ /\ litre$

$\therefore \quad a = 0.4 mole = 0.4\times 53.5 = 21.4g\ /\ litre$

Thus, $[Salt] = 0.4\ M$ *and* $[Base] = 0.2\ M$

14. **B**

$\because \quad \Delta G = \Delta H - T\Delta S$

For a non-spontaneous reaction ΔG = +ve;

$\therefore \quad \Delta H = T\Delta S = +ve$

or $\quad \Delta H > T\Delta S \quad or \quad T < (\Delta H / \Delta S)$

$\because \quad \Delta S = 70 JK^{-1}mol^{-1}$

Thus, $\quad T < \left[\left(30\times10^{3}\right)/70\right]$

$T < 428.57\ K$

15. **B**

16. **C**

Two possible isomers of C_4H_9Br are:

$CH_3—CH_2—CH_2—CH_2—Br$ $\quad$ $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—Br$

[I] $\quad$ [II]

(1°) $\quad$ (1°)

According to the question, compound (A) on reaction with sodium does not give the same product produced by n-butylbromide. So, (A) cannot be [I]

$2CH_3—CH_2—CH_2—CH_2—Br + 2Na \xrightarrow{\text{Ether}}$

n-butylbromide

$CH_3—CH_2—CH_2—CH_2—CH_2—CH_2—CH_2—CH_3$

n-octane (not formed in the reaction)

C_8H_{18}

Now second must be the correct isomer.

$CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—Br \xrightarrow{KOH(alc.)}$

Iso-butylbromide

(1-bromo-2-methylpropane) (A)

$CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}=CH_2 + KBr + H_2O \xrightarrow{HBr} CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle Br}{\overset{|}{\underset{|}{C}}}}—CH_3$

2-methylpropene $\quad$ 2-bromo-2-methyl propane

(B) $\quad$ (C)

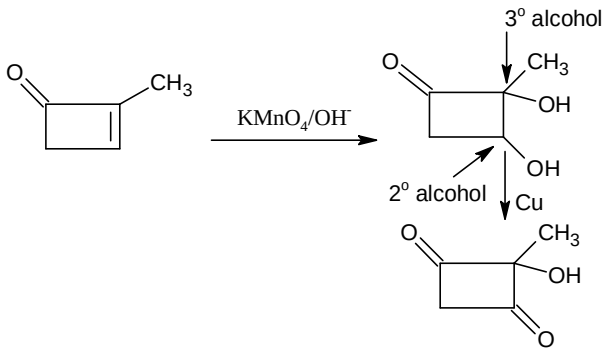
$2CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—Br + 2Na \xrightarrow[\text{Wurtz reaction}]{\text{Dry ether}}$

$CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH_2}}—CH_2—CH_2—\underset{\displaystyle CH_3}{\underset{|}{CH_2}}—CH_3$

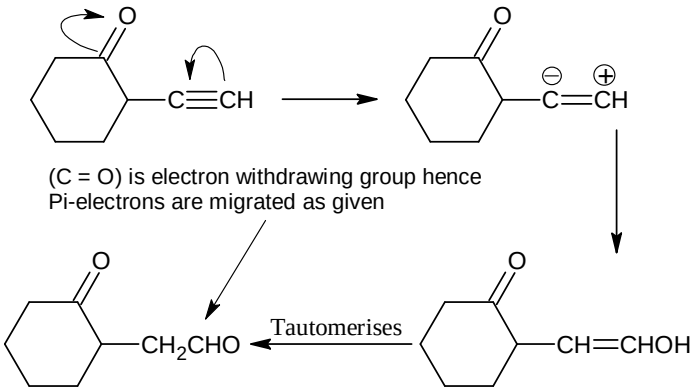
2,5-dimethylhexane

(C_8H_{18})D

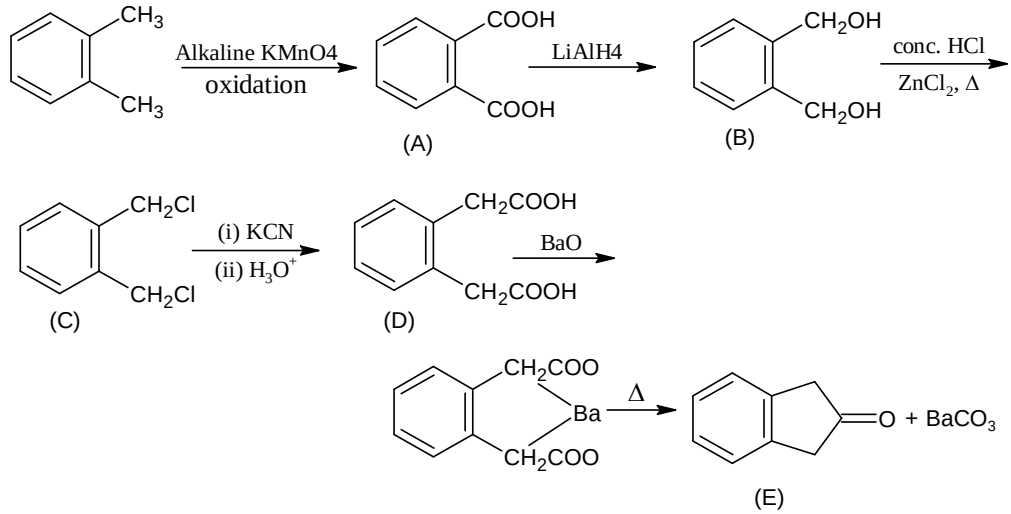
17. **B**



18. **B**



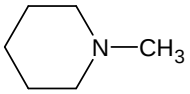
19. **A**



20. **B**

21. **D**

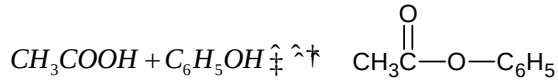
22. **C**



No H-atom at N-atm (3° amine) hence, no H-bonding. Thus least boiling point

23. **B**

Compound (B) gives violet colour with neutral $FeCl_3$, thus, it is phenol. Compound (C) Gives blood red colour with neutral $FeCl_3$. Thus, it is CH_3COOH

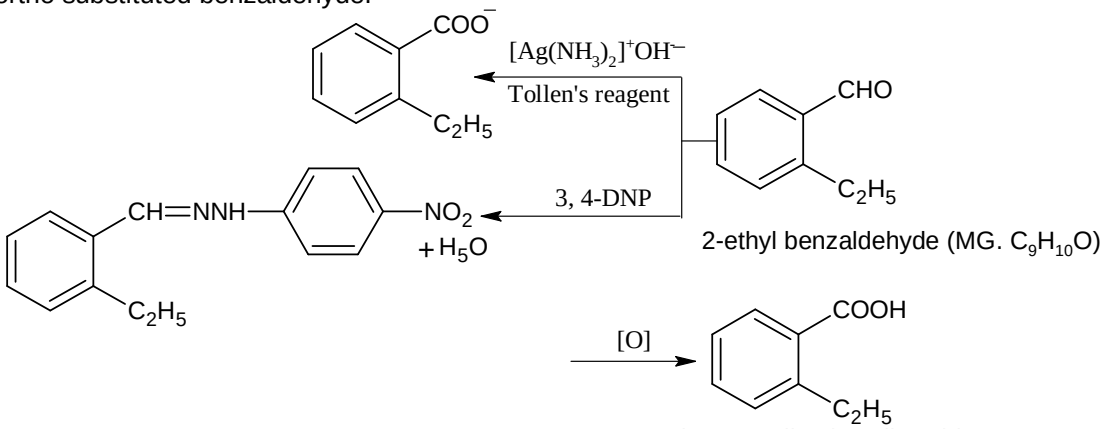


24. **A**

(i) Since, the given compound with MF $C_9H_{10}O$ forms a 2,4-DNP derivative and reduces Tollen's reagent, it must be an aldehyde.

(ii) Since, it undergoes cannizzaro reaction, therefore, CHO group is directly attached to the benzene ring.

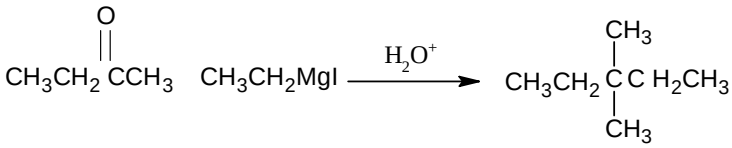
(iii) Since, on vigorous oxidation, it gives 1, 2-benzene dicarboxylic acid, there fore, it must be an ortho substituted benzaldehyde.



25. **B**

26. **C**

27. **D**



3° alcohol

28. **D**

29. **A**

30. **D**

Write the reaction of acid & base to get the answer.

Mathematics

PART – III

SECTION – A

1. **A**

$$f(x)=\begin{cases}x^3-x^2+x-2 & x>2\\ x^3-x^2+2-x & 1<x\le 2\\ x^2-x^3+2-x & 0<x\le 1\\ x^3-x^2+2-x & x\le 0\end{cases}$$

$$f'(x)=\begin{cases}3x^2-2x+1 & x>2\\ 3x^2-2x-1 & 1<x\le 2\\ 2x-3x^2-1 & 0<x\le 1\\ 3x^2-2x-1 & x\le 0\end{cases}$$

$f'(x)=0 \quad 3x^2-2x+1=0 \quad \Rightarrow x=\phi, \quad X>2$

$f'(x)=0 \quad 3x^2-2x-1=0 \Rightarrow x=\dfrac{2\pm 4}{6} \Rightarrow x=1,\dfrac{-1}{3}; 1<x<2 \Rightarrow x=\phi$

$f'(x)=0 \quad 2x-3x^2-1=0 \quad 0 < x \le 1$

$\Rightarrow 3x^2-2x+1=0 \quad \Rightarrow x=\phi$

$f'(x)=0 \quad 3x^2-2x-1=0 \quad x \le 0$

$\Rightarrow x=1,\dfrac{-1}{3} \quad \Rightarrow x=-1/3$

Also $x < 1 \quad \Rightarrow f'(x)<0 \quad x > 1 \quad \Rightarrow f'(x)>0$

$\therefore$ minimum at x = 1.

2. **C**

$f(x)=x^3+\int_1^x f(t)\,dt$

$xf'(x)+f(x)=3x^2+f(x)$

$f'(x)=3x$

$f(x)=\dfrac{3x^2}{2}+c$ .....(i) Also, $f(1)=1^2+\dfrac{1}{1}\int_1^1 f(t)\,dt=1$

$\therefore \quad f(x)=\dfrac{3x^2-1}{2}$

3. **C**

Since $f'(x)-2f(x)>0, \quad x\ge 1$

$\Rightarrow \dfrac{d}{dx}\left(e^{-2x}\cdot f(x)\right)>0, \quad x\ge 1$

$h(x) = e^{-2x} f(x)$ is increasing in x ≥ 1. Now x > 1

$\Rightarrow e^{-2x} f(x) > e^{-2} \cdot f(1)$

$\Rightarrow f(x) > e^{2x-2}, \quad x > 1$

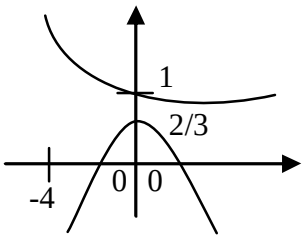
4. **A**

Since $A(adj(A)) = |A| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ hence K = |A| = 1

5. **A**

$\therefore f'(x) = x^2 e^x + 2x \cdot e^x - 2e^x$

$= e^x (x^2 + 2x - 2)$



$f''(x) = x^2 e^x + x4e^x = xe^x (x+4)$

$= x^2 e^x + 4 \times e^x \; xe^x (x+4) < 0.$

Also f(x) > 0 and f(0) > 0

Hence f(x) > 0 $\forall x \in (-4, 0)$

Hence no. real roots.

$(x^2 - 2)e^x = -3 \qquad \Rightarrow x^2 - 2 = -3e^{-x} \Rightarrow \frac{2 - x^2}{3} = e^{-x}$

6. **D**

$f(x) = e^x - 3x^2$

$f'(x) = e^x - 6x$

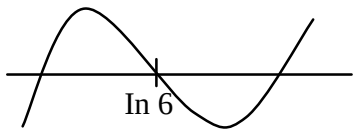
$f''(x) = e^x - 6$

$f''(x) > 0$

$\Rightarrow e^x > 6$

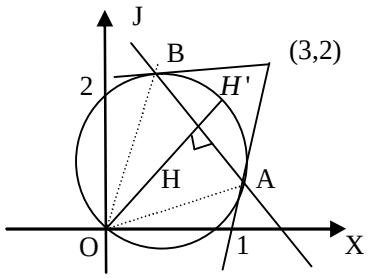
$\Rightarrow x > \ln 6.$

$f''(x) < 0 \Rightarrow e^n < 6 = x < \ln 6$



7. **B**

If altitude from O meet the circle at H' then ortho centre of triangle OAB is the mirror image of H' about the line AB.



$3x+2y-(x+3)-(y+2)=0$

$\Rightarrow 2x+y=5 \quad \Rightarrow$ AB

$y=-2$ 1/ 2 $y=\dfrac{1}{2} \Rightarrow$ x = 2y (OMA')

8. **C**

Let $I_2=\int_0^1\left(1-x^n\right)^{m+1}$

$$I_2=\int_0^1\left(1-x^n\right)^m-\int_0^1 x^n\left(1-x^n\right)^m$$

$$I_2=I_1-\left\{\left(x\frac{\left(1-x^n\right)^{m+1}}{n(m+1)}\right)_0^1-\int_0^1\frac{\left(1-x^n\right)^{m+1}}{n(m+1)}\right\}$$

$I_2 = I_1 + \dfrac{I_2}{n(m+1)}$

9. **B**

Now $\dfrac{dy}{dx}=2x=2 \qquad \Rightarrow x=1 \qquad \Rightarrow y=3$

Hence point is (1, 3)

10. **B**

Let z = x + iy

$x^2-y^2+2xiy=ix-y+\sqrt{x^2+y^2}-2$

Now $x^2-y^2=-y+\sqrt{x^2+y^2}-2$ ....(i)

Also $2xy-x=0 \qquad \Rightarrow$

x = 0 and $y=1/2$

Now x = 0, $-y^2=-y+|y|-2$

$-y^2=-2$ y >0

$y=\pm\sqrt{2} \qquad \Rightarrow y=\sqrt{2}$

$\left(0,\sqrt{2}\right)$

Also $y\le 0.$

$-y^2=-2y-2$

$y^2-2y-2=0$

$(y-1)^2=3 \quad \Rightarrow y=\pm\sqrt{3}$. hence $\left(0,1-\sqrt{3}\right)$

Now $y=1/2$

$x^2-1/4=-1/2-2+\sqrt{x^2+1/4}$

$x^2+\frac{9}{4}=\sqrt{x^2+1/4}$, Let x^2 = t

$t^2+\frac{81}{16}+\frac{9t}{2}=t+\frac{1}{4} \qquad 16t^2+56t+77=0$

Equation has no any real solution.

11. **B**

$y\,dx-x\,dy+3x^2y^2\cdot e^{x^3}dx=0$

$\Rightarrow \frac{y\,dx-x\,dy}{y^2}+3x^2\cdot e^{x^3}dx=0$

$\Rightarrow d\left(\frac{x}{y}\right)+d\left(e^{x^3}\right)=0$

$\Rightarrow \frac{x}{y}+e^{x^3}=C$

12. **B**

$h(x)=\frac{1}{2}\frac{d}{dx}\left(x^2-1\right)\left(f'(x)\right)^2$

Hence $f(x)$ have 4 zeros

hence 10.

13. **A**

Roots are $\alpha_1,1-\alpha_1,\alpha_2,1-\alpha_2,\alpha_3,1-\alpha_3,\frac{1}{2},\frac{1}{2}$

Sum = $3+\frac{1}{2}+\frac{1}{2}=4$

14. **B**

$g\left(f(x)\right)=x \Rightarrow g'\left(f(x)\right)f'(x)=1$

$g'\left(f(0)\right)=\frac{1}{f'(0)}=1/2=g'(-4)$

Now $f'\left(1/2\right)=5\times\frac{1}{16}+2=\frac{37}{16}$

$f\left(g(x)\right)=x$

$\Rightarrow f'\left(g(x)\right)g'(x)=1$

15. **A**

Equ^{n} of circle is $(x-1)^2+(y-2)^2+\lambda(y-2x)=0$

Put y = 0

$x^2+1-2x+4-2\lambda x=0$

$x^2-2x(1+\lambda)+5=0$

Now D= 0

$4(1+\lambda)^2-20=0$

$(1+\lambda)^2=5$

$\lambda=\pm\sqrt{5}-1$

$x^2+y^2-2x-4y+(\sqrt{5}-1)y-2(\sqrt{5}-1)x=0$

$x^2+y^2-2x\sqrt{5}-y(5-\sqrt{5})+5=0$

16. **D**

$$\overrightarrow{a_1}=(\vec{a}\cdot\vec{b})\vec{b}=\frac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}$$

$$\Rightarrow \overrightarrow{a_2}=\vec{a}-\overrightarrow{a_1}=\vec{a}-\frac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}$$

Now, $\overrightarrow{a_1}\times\overrightarrow{a_2}\dfrac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}\times\left(\vec{a}-\dfrac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}\right)$

$$=\frac{(\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a})}{|\vec{b}|^2}$$

17. **A**

$|\vec{a}+\vec{b}|=|-\sqrt{3}\vec{c}|$

$(\vec{a})^2+(\vec{b})^2+2\vec{a}\cdot\vec{b}=3|\vec{c}|^2$

$2\vec{a}\cdot\vec{b}=3|\vec{c}|^2$

$2\vec{a}\cdot\vec{b}=1$

$\vec{a}\cdot\vec{b}=1/2\Rightarrow\cos\theta=1/2\Rightarrow\theta=\dfrac{\pi}{3}$

18. **B**

$\left((\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c})\right)\cdot(\vec{c}\times\vec{a})$

$\left(\left[\vec{a}\vec{b}\vec{c}\right]\vec{b}\cdot(\vec{c}\times\vec{a})\right)$

$= \left[\vec{a}\vec{b}\vec{c}\right]^2 = 16$

19. **A**

$$\frac{f(x)}{1+x} = 1 + \int_1^x \frac{f^2(x)}{1+t}dt$$

$$\frac{(1+x)f'(x) + f(x)}{(1+x)} = \frac{f^2(x)}{(1+x)}$$

$$(1+x)\frac{dy}{dx} + y = y^2$$

$$(1+x)\frac{dy}{dx} = y^2 - y$$

$$\frac{dy}{y^2 - y} = \frac{dx}{1+x}$$

$$\int\left(\frac{1}{y-1} - \frac{1}{y}\right)dy = \ln(1 + x) + c$$

$$ln\left(\frac{y-1}{y}\right) = ln(1+x) + c$$

$$ln\left(\frac{1}{2}\right) = ln(2) + c$$

$$\Rightarrow c = -2ln2 = \ln(1/4)$$

$$\therefore \frac{y-1}{y} = \left(\frac{1+x}{4}\right)$$

$$4y - 4 = y(1+x) \Rightarrow y(3-x) = 4$$

$$y = \frac{4}{3-x}$$

$$\therefore f(7) = \frac{4}{3-7} = -1$$

20. **A**

$$\left(\frac{r}{r+1}\right){}^nC_r = \left(1 - \frac{1}{r+1}\right){}^nC_r$$

$$= {}^nC_r - \frac{1}{r+1}\ {}^nC_r$$

On solving we get $\frac{1}{n+1}$

21. **A**

Since $a_1 + a_{11} = a_2 + a_{10} = a_3 + a_9 = a_4 + a_8$

$a_5 + a_7 = 20$

22. **A**

23. **C**

Since $\left|\frac{z_1 - z_2}{z_1 + z_2}\right| = 1 \qquad \Rightarrow \left|\frac{z_1}{z_2} - 1\right| = \left|\frac{z_1}{z_2} + 1\right|$

$\therefore \frac{z_1}{z_2}$ lies on y – axis.

$\therefore \left|\arg\left(\frac{z_1}{z_2}\right)\right| = \pi / 2$

24. **B**

Left Z = 2(cosθ + i sin θ)

$z = 2\left(\cos\theta + i\sin\theta\right) + \frac{1}{2}\left(\cos\theta - i\sin\theta\right)$

w = $h + ik = \frac{5\cos\theta}{2} + \frac{3}{2} i\sin\theta$

$\cos\theta = \frac{2h}{5}, \sin\theta = \frac{2k}{3}$

Hence locus is $\frac{4x^2}{25} + \frac{4y^2}{9} = 1$

$b^2 = a^2\left(1 - e^2\right)$

$\Rightarrow \frac{9}{4} = \frac{25}{4}\left(1 - e^2\right)$

$e = 4/5$

25. **A**

$T_n = \frac{1}{2^{n-1} + 2^{-n} - 1.5} = \frac{2^n}{2^{2n-1} - 1.5\ 2^n + 1}$

$T_n = \frac{2^{n+1}}{4^n - 3.2^n + 2} = \frac{2^{n+1}}{\left(2^n - 2\right)\left(2^n - 1\right)} = \frac{2^n}{\left(2^{n-1} - 1\right)\left(2^n - 1\right)} = \frac{2^n}{2^{n-1} - 1} - \frac{2^{n+1}}{2^n - 1}$

$T_n = V_n - V_{n+1}$

$T_1 = V_1 - V_2$

$T_2 = V_2 - V_3$

$T_n = V_n - V_{n+1}$

$S_n = V_2 - V_{n+1}$

$S_n = 4 - \frac{2^{n+1}}{2^n - 1} = 4 - \frac{2^{50}}{2^{49} - 1}$

26. **A**

$\int_0^m \{2x\} e^{\{2x\}} dx + 200 \int_0^{1/2} (2x) e^{2x} dx$

$$2\int_0^m xe^{2x}dx + 400\int_0^{1/2} xe^{2x}dx$$

$$2\left\{\frac{xe^{2x}}{2} - \frac{1}{4}e^{2x}\right\}_0^m + 400\left\{\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x}\right\}_0^{1/2}$$

$$= me^{2m} - \frac{1}{2}e^{2m} + \frac{1}{2} + 400\left\{\frac{1}{4}e - \frac{1}{4} + e\frac{1}{4}\right\}$$

$$e^{2m}\left(m - \frac{1}{2}\right) + \frac{1}{2} + 100$$

$$\frac{e^{2m}(2m-1) + 201}{2}$$

27. **A**

Any tangents will be in the form y = mx + c, where $c^2 = 36m^2 - 9$
If it passes through (0, 4), then 4 = c

$$\Rightarrow 16 = 36m^2 - 9$$

$$\Rightarrow m = \pm\frac{5}{6}.$$

Hence tangent are $6y = 5x + 24, 6y = -5x + 24$

28. **C**

The given distance is clearly the length of semi major axis

Thus , $\sqrt{\frac{a^2 + 2b^2}{2}} = a$

$$\Rightarrow 2b^2 = a^2 \qquad \Rightarrow 2a^2\left(1 - e^2\right) = a^2$$

$$e^2 = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}}$$

29. **D**

Given m + n = 100
Also m.n > 1600
Or m (100 – n) > 1600 on solving
20 , m < 80 $\Rightarrow$ 21 $\leq$ m $\leq$ 79
Hence total number of favorable ways is 79 – 21 + 1 = 59 hence required probability is 59/99.

30. **D**

$$2^{(x+1)^2+2} \leq 4 - \left(x^2 - 1\right)^2$$

Since $4 - \left(x^2 - 3\right)^2 \leq 4$ and $2^{(x+1)^2+2} \geq 4$

hence $2^{(x+1)^2+2} = 4$ and $4 - \left(x^2 - 1\right)^2 = 4$

$\left(x^2 - 1\right)^2 = 0 \quad \Rightarrow x = \pm 1$ and x = - 1 hence x = -1

$$\lim_{x\to -1}\left(1 + \sin\pi x\right)^{\tan\frac{x\pi}{2}} = \lim_{x\to -1} e^{\sin\pi x\left(\frac{\sin\frac{\pi x}{2}}{\cos\frac{\pi x}{2}}\right)} = e^{-2}$$